

Optimal investment under transient price impact

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The time profile of price impact

How long does the effect of a trade persist?

instantaneous



permanent

Instantaneous impact

- additional cost when executing, instantly disappears after transaction
- cost only depends on current traded quantity
- market heals immediately

Permanent impact

- every trade shifts the price permanently
- decision made today changes all subsequent prices
- there is no resilience in the market

Frictionless market; expected utility

P_t : mid-price, X_t : risky position, $H_t := X_t - X_{t-1}$, ξ_t : cash account.

Textbook frictionless model

$$\xi_t - \xi_{t-1} = -P_t(X_t - X_{t-1}) = -P_t H_t.$$

If $X_0 = 0$ and $X_T = 0$, then

$$\xi_T^X = \xi_0 - \sum_{t=1}^T P_t(X_t - X_{t-1}) = \xi_0 + \sum_{t=1}^{T-1} X_t(P_{t+1} - P_t).$$

Expected utility

$$u : \mathbb{R} \rightarrow \mathbb{R}, \quad X \mapsto \mathbb{E}[u(\xi_T^X)].$$

Instantaneous price impact

FRICTION

Notation: $H_t = X_t - X_{t-1}$.

$$\xi_t^X - \xi_{t-1}^X = -P_t H_t - G_t(H_t) = -(P_t + G_t(H_t)/(H_t))(H_t).$$

What do we know about G_t ?

- $G_t(\omega, h)$ is measurable and non-negative; the friction may be random and time-dependent.
- $h \mapsto G_t(\omega, h)$ is convex.
- Inactivity is cheapest: $G_t(\omega, h) \geq G_t(\omega, 0)$.
- For large trades, the cost grows superlinearly: $G_t(h)$ grows faster than $|h|$.

Typical example: $G_t(h) = \lambda|h|^\alpha$, $\alpha > 1$, $\lambda > 0$.

CONVEX ANALYSIS

Permanent price impact

IMPACT THROUGH INVENTORY

- the affected price is determined by the trader's current inventory: $P_t^X = F_t(X_t)$
- inventory X_t contains the cumulative effect of all previous trades.
- once the position has changed, the affected price level remains shifted

TWO MODELLING SCALES

- **General: semimartingale family**

$$F_t(x) = \tilde{P}_t^x, \quad x \in \mathbb{R},$$

where the inventory x selects the price process $(\tilde{P}^x)_{x \in \mathbb{R}}$. Bank–Baum (2004)

- **Simple: linear impact**

$$F_t(x) = P_t + \iota x, \quad \iota \geq 0.$$

This specification can also be motivated by utility-indifference pricing. Bank–Kramkov (2015)

The time profile of price impact

How long does the effect of a trade persist?

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The truth is in between.

What's not instantaneous and not permanent?

The time profile of price impact

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Discrete time: transient price impact

Primitives.

- Price and liquidity:

$$P = (P_t)_{t=1}^T$$

mid-price, adapted

$$\delta = (\delta_t)_{t=1}^T$$

depth, adapted, $\delta_t > 0$

$$r = (r_t)_{t=0}^{T-1}$$

resilience, adapted, $r_t \geq 0$

$$X = (X_t)_{t=0}^T$$

inventory, predictable

SPREAD DYNAMICS

$$\zeta_0^X = \zeta_0 \geq 0, \quad \zeta_t^X = e^{-r_{t-1}} \zeta_{t-1}^X + \frac{|X_t - X_{t-1}|}{\delta_t}, \quad t = 1, \dots, T.$$

CASHFLOW DYNAMICS

$$\xi_0^X = 0, \quad \xi_t^X - \xi_{t-1}^X = -P_t(X_t - X_{t-1}) - \zeta_t^X |X_t - X_{t-1}|, \quad t = 1, \dots, T.$$

Closed-form expressions

RESILIENCE FACTOR

$$\rho_{j,t} := \exp\left(-\sum_{i=j}^{t-1} r_i\right), \quad 0 \leq j \leq t \leq T; \quad \rho_{t,t} = 1.$$

SPREAD

$$\zeta_t^X = \rho_{0,t} \zeta_0 + \sum_{j=1}^t \rho_{j,t} \frac{|X_j - X_{j-1}|}{\delta_j}, \quad t = 1, \dots, T.$$

TERMINAL VALUE

$$\xi_T^X = \zeta_0 - \sum_{t=1}^T P_t (X_t - X_{t-1}) - \sum_{t=1}^T |X_t - X_{t-1}| \left(\rho_{0,t} \zeta_0 + \sum_{j=1}^t \rho_{j,t} \frac{|X_j - X_{j-1}|}{\delta_j} \right).$$

Continuous time: transient price impact

Primitives.

Price, liquidity and inventory:

$$P_t^X = P_t + \iota X_t, \quad \iota \geq 0 \quad \delta_t > 0 : \text{market depth} \quad r_t \geq 0 : \text{resilience}$$

$$X_t = x_0 + X_t^\uparrow - X_t^\downarrow, \quad d|X|_t := dX_t^\uparrow + dX_t^\downarrow.$$

SPREAD DYNAMICS

$$d\zeta_t^X = \frac{1}{\delta_t} d|X|_t - r_t \zeta_t^X dt, \quad \zeta_{0-}^X = \zeta_0.$$

CASHFLOW

$$\xi_t^X = \xi_0 - \int_{[0,t]} P_s^X \circ dX_s - \int_{[0,t]} \zeta_s^X \circ d|X|_s.$$

The optimization problem

ADMISSIBLE STRATEGIES

$\mathcal{X}$: set of admissible strategies, $X \in \mathcal{X}$.

ξ_T^X : terminal wealth generated by X .

QUESTION

Does there exist $X^* \in \mathcal{X}$ such that

$$\mathbb{E}\left[u\left(\xi_T^{X^*}\right)\right] = \sup_{X \in \mathcal{X}} \mathbb{E}\left[u\left(\xi_T^X\right)\right]?$$

Bank, Voß (2018)

- δ, r constant;
- $u \in C^1$, strictly concave, increasing, bounded from above;
- existence and uniqueness of an optimal strategy.

Bank, Dolinsky (2019)

- δ_t, r_t stochastic;
- u strictly concave, increasing, differentiable;
- shadow-price based verification theorem for optimality of a candidate strategy;
- no general existence result;
- **structural restriction: restores convexity.**

The monotonicity condition

THE RATIO

$$o_t := \frac{\delta_t}{\rho_t^2}, \quad \rho_t := \exp\left(\int_0^t r_s ds\right).$$

Bank–Dolinsky monotonicity criterion

o_t is strictly decreasing.

INTERPRETATION

- the ratio links the paths of market depth and resilience;
- for stochastic δ_t and r_t , monotonicity is not automatic.

THE RAISED QUESTION

Can we remove this monotonicity condition?

SUPERHEDGING SET

$$S := \left\{ Y \in L^0 : \exists X \in A_0 \text{ such that } \xi_T^X \geq Y \text{ a.s.} \right\}.$$

CONVEX STRUCTURE AND DUALITY

- monotonicity of δ_t/ρ_t^2 *implies* convex liquidity costs;
- convex liquidity cost means concave terminal wealth;
- with a convex strategy set, this gives convexity of S .
- Komlós: closedness of the superhedging set;
- Hahn–Banach: separation from S ;
- separation gives the dual representation of $\pi(H)$;
- the dual system yields shadow-price verification for utility maximization.

... Bank, Voß (2018) takes a similar Komlos-compactness route.

Non-convexity: the setup

EXAMPLE

$$T = 3, \quad r_0 = r_1 = r_2 = 0, \quad P_0 = \zeta_0 = 0, \\ P_1, P_2, P_3 \sim \mathcal{N}(0, 1) \text{ independent}, \quad \delta_1 = 1, \quad \delta_2 = \delta_3 = 10.$$

Roundtrip condition:

$$X_0 = X_3 = 0 \quad \Longleftrightarrow \quad H_1 + H_2 + H_3 = 0.$$

SUPERHEDGING SET

$$S := \left\{ Y \in L^0 : \exists X \in A_0 \text{ such that } \xi_T^X \geq Y \text{ a.s.} \right\}.$$

Claim: S is not convex.

Non-convexity: what convexity would imply

Take two deterministic roundtrips:

$$H = (1, 1, -2), \quad G = \left(\frac{3}{2}, 0, -\frac{3}{2}\right).$$

PAYOFFS IN THE SUPERHEDGING SET

Since H and G are admissible,

$$\xi_T^H \in S, \quad \xi_T^G \in S.$$

IF S WERE CONVEX

There would exist some roundtrip $\bar{H}$ such that

$$\xi_T^{\bar{H}} \geq \frac{\xi_T^H + \xi_T^G}{2} \quad \text{a.s.}$$

RANDOM PRICE TERMS

Independence of P_1, P_2, P_3 forces

$$\bar{H} = \frac{H + G}{2} = \left(\frac{5}{4}, \frac{1}{2}, -\frac{7}{4}\right).$$

Non-convexity: where the contradiction appears

For any roundtrip K ,

$$\xi_T^K = - \sum_{t=1}^T P_t K_t - C(K),$$

where $C(K)$ is the liquidity-cost part.

AFTER THE PRICE TERMS CANCEL

The inequality

$$\xi_T^{\bar{H}} \geq \frac{\xi_T^H + \xi_T^G}{2}$$

reduces to

$$C(\bar{H}) \leq \frac{C(H) + C(G)}{2}.$$

BUT THE COST INEQUALITY FAILS

$$C(\bar{H}) = 4.79375, \quad \frac{C(H) + C(G)}{2} = 4.7125.$$

Hence

$$C(\bar{H}) > \frac{C(H) + C(G)}{2}.$$

Main result: existence beyond convexity

Assumptions.

- Utility:

$u : \mathbb{R} \rightarrow \mathbb{R}$ non-decreasing, continuous, bounded from above, $\lim_{x \rightarrow -\infty} u(x) = -\infty$.

- Price, liquidity and strategy:

$P = (P_t)_{t=0}^T$ mid-price, adapted $r = (r_t)_{t=0}^{T-1}$ resilience, adapted, $r_t \geq 0$ $\delta = (\delta_t)_{t=1}^T$ depth, adapted, $\delta_t \geq \underline{\delta} > 0$ $X = (X_t)_{t=0}^T$ strategy, predictable

- Liquidation constraint:

$$A_0 := \{X \in A : X_0 = 0, X_T = 0\}.$$

TRANSIENT DYNAMICS

$$\zeta_0^X = \zeta_0 \geq 0, \quad \zeta_t^X = e^{-r_{t-1}} \zeta_{t-1}^X + \frac{|X_t - X_{t-1}|}{\delta_t}, \quad t = 1, \dots, T.$$

$$\xi_0^X = 0, \quad \xi_t^X - \xi_{t-1}^X = -P_t(X_t - X_{t-1}) - \zeta_t^X |X_t - X_{t-1}|, \quad t = 1, \dots, T.$$

RESULT

There exists $X^* \in A_0$ such that

$$\mathbb{E}[u(\xi_T^{X^*})] = \sup_{X \in A_0} \mathbb{E}[u(\xi_T^X)].$$

Outline of proof – indirect utility

ONE-STEP CASHFLOW

For a trading history $h_1, \dots, h_t$, define

$$\kappa_t(h_1, \dots, h_t) := -P_t h_t - \zeta_t(h_1, \dots, h_t) | h_t|.$$

Then the cash account satisfies

$$\xi_t^X = \xi_{t-1}^X + \kappa_t(H_1, \dots, H_t).$$

FIRST BACKWARD STEP

At time $T - 1$, keep the current cash x and the past trades

$$h_1, \dots, h_{T-1}$$

as state variables. We denote by H the last admissible choice.

Define the conditional indirect utility

$$J_{T-1}(x, h_1, \dots, h_{T-1}) := \operatorname{ess\,sup}_H \mathbb{E}[u(x + \kappa_T(h_1, \dots, h_{T-1}, H)) | \mathcal{F}_{T-1}].$$

Outline of proof – indirect utility

CONDITIONAL INDIRECT UTILITY

For fixed current cash x and past trades $h_1, \dots, h_{T-1}$, we defined

$$J_{T-1}(x, h_1, \dots, h_{T-1}) := \operatorname{ess\,sup}_H \mathbb{E}[u(x + \kappa_T(h_1, \dots, h_{T-1}, H)) \mid \mathcal{F}_{T-1}].$$

MEASURABLE OPTIMAL SELECTOR

The one-step argument shows that this essential supremum is attained: there exists a measurable decision rule

$$H_T^* = H_T^*(x, h_1, \dots, h_{T-1})$$

such that

$$J_{T-1}(x, h_1, \dots, h_{T-1}) = \mathbb{E}[u(x + \kappa_T(h_1, \dots, h_{T-1}, H_T^*)) \mid \mathcal{F}_{T-1}].$$

The last step is now solved and packaged into J_{T-1} .

Proof idea: one step further back

SECOND BACKWARD STEP

For fixed current cash x and past trades

$$h_1, \dots, h_{T-2},$$

define

$$J_{T-2}(x, h_1, \dots, h_{T-2}) := \operatorname{ess\,sup}_H \mathbb{E} \left[J_{T-1} \left(\underbrace{x + \kappa_{T-1}(h_1, \dots, h_{T-2}, H)}_{\text{new cash position}}, \underbrace{h_1, \dots, h_{T-2}, H}_{\text{updated history}} \right) \middle| \mathcal{F}_{T-2} \right].$$

WHAT CHANGED?

- The utility u is no longer visible directly:

$$u \rightsquigarrow J_{T-1}.$$

- The control H enters through two channels:
 - it changes the current cash position;
 - it becomes part of the trading history for future steps.

Proof idea: backward values, forward strategy

BACKWARD CONSTRUCTION

The backward induction gives value functions and decision rules:

$$u \longrightarrow (J_{T-1}, H_T^*) \longrightarrow (J_{T-2}, H_{T-1}^*) \longrightarrow \cdots \longrightarrow (J_0, H_1^*).$$

1ST STEP, $T = 1$

$$\hat{x}_0 := 0,$$

$$\hat{H}_1 := H_1^*(\hat{x}_0),$$

$$\hat{x}_1 := \hat{x}_0 + \kappa_1(\hat{H}_1).$$

2ND STEP, $T = 2$

$$\hat{H}_2 := H_2^*(\hat{x}_1, \hat{H}_1),$$

$$\hat{x}_2 := \hat{x}_1 + \kappa_2(\hat{H}_1, \hat{H}_2).$$

$T = 3$, CONTINUE

$$\hat{H}_3 := H_3^*(\hat{x}_2, \hat{H}_1, \hat{H}_2),$$

$$\hat{x}_3 := \hat{x}_2 + \kappa_3(\hat{H}_1, \hat{H}_2, \hat{H}_3),$$

$$\vdots$$

The value functions are constructed backward; the strategy is constructed forward.

Proof idea: what makes one step work

A KEY BOUND

For fixed past trades, the current trade is quadratically penalized:

$$\kappa_t(h_1, \dots, h_t) \leq -P_t h_t - \frac{h_t^2}{\delta_t}.$$

$|h_t|$ large $\implies$ too expensive to be optimal.

Thus the one-step search can be restricted to a random bounded set.

ONE-STEP OUTPUT

Suppose that the continuation value J_t is already constructed.

The one-step argument constructs J_{t-1} and a measurable selector H_t^* such that:

- the essential supremum defining J_{t-1} is attained by H_t^* ;
- J_{t-1} is continuous in the state variables;
- J_{t-1} is non-decreasing in the cash variable;
- the same optimality identity remains valid after substituting random states.

Proof idea: concavity is not restored

A ONE-STEP WARNING

Consider a one-step frictionless model with admissible strategies

$$\Phi = \{0, 1\},$$

and utility

$$u(x) = \log x, \quad x > 0.$$

Let the return R satisfy

$$\mathbb{P}(R = 2) = \mathbb{P}(R = -1) = \frac{1}{2}.$$

INDIRECT UTILITY

For initial capital $z > 1$,

$$\begin{aligned} \bar{u}(z) &:= \sup_{\phi \in \Phi} \mathbb{E}[u(z + \phi R)] \\ &= \max \left\{ \log z, \frac{\log(z + 2) + \log(z - 1)}{2} \right\}. \end{aligned}$$

This function is not concave.

